

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Touch Display
Manufacturer: Avocor
Firmware Version: V1.14
Model(s): AVF-6510, AVF-8410, VTF-6510, VTF-8410

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0001-b003	1.4.1

Version History

Module Version	Date	Notes
1_1_0_0	10/31/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='AVF-6510')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='AVF-6510')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='AVF-6510')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Native' 'Letterbox'	Value 'Full Screen'	Value '4:3'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Native')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage ¹	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Input	Value 'VGA' 'HDMI 3' 'DisplayPort'	Value 'HDMI 1' 'HDMI 4' 'IPC/OPS'	Value 'HDMI 2' 'HDMI 5 (Front)' 'Media Player (Win/Android)'
# Input example InterfaceName.Set('Input', 'VGA')			
Command MenuNavigation	Value 'Menu' 'Left' 'Exit'	Value 'Up' 'Right' 'Info'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command PictureMode	Value 'Cinema' 'User'	Value 'Sport' 'Game'	Value 'Vivid'

# PictureMode example InterfaceName.Set('PictureMode', 'Cinema')			
Command PIP	Value 'Off' 'Large' '4 Windows'	Value 'Small' 'Side by Side'	Value 'Medium' '3 Windows'
# PIP example InterfaceName.Set('PIP', 'Off')			
Command PIPIInput	Value 'VGA' 'HDMI 3' 'DisplayPort'	Value 'HDMI 1' 'HDMI 4' 'IPC/OPS'	Value 'HDMI 2' 'HDMI 5 (Front)' 'Media Player (Win/Android)'
# PIPIInput example InterfaceName.Set('PIPIInput', 'VGA')			
Command PIPIInputWindow2	Value 'VGA' 'HDMI 3' 'DisplayPort'	Value 'HDMI 1' 'HDMI 4' 'IPC/OPS'	Value 'HDMI 2' 'HDMI 5 (Front)' 'Media Player (Win/Android)'
# PIPIInputWindow2 example InterfaceName.Set('PIPIInputWindow2', 'VGA')			
Command PIPIInputWindow3	Value 'VGA' 'HDMI 3' 'DisplayPort'	Value 'HDMI 1' 'HDMI 4' 'IPC/OPS'	Value 'HDMI 2' 'HDMI 5 (Front)' 'Media Player (Win/Android)'
# PIPIInputWindow3 example InterfaceName.Set('PIPIInputWindow3', 'VGA')			
Command PIPPosition	Value 'Bottom Left' 'Top Right'	Value 'Bottom Right'	Value 'Top Left'
# PIPPosition example InterfaceName.Set('PIPPosition', 'Bottom Left')			
Command PIPSwap	Value None		
# PIPSwap example InterfaceName.Set('PIPSwap', None)			
Command Power	Value 'On'	Value 'Off'	
# Power example			

InterfaceName.Set('Power', 'On')			
Command RCEmulation	Value 'Freeze' 'Volume Down'	Value 'Blank Screen'	Value 'Volume Up'
# RCEmulation example InterfaceName.Set('RCEmulation', 'Freeze')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

¹ For VGA input only

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Native' 'Letterbox'	Value 'Full Screen'	Value '4:3'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command Input	Value 'VGA' 'HDMI 3'	Value 'HDMI 1' 'HDMI 4'	Value 'HDMI 2' 'HDMI 5 (Front)'

	'DisplayPort'	'IPC/OPS'	'Media Player (Win/Android)'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command PictureMode	Value 'Cinema' 'User'	Value 'Sport' 'Game'	Value 'Vivid'
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command PIP	Value 'Off' 'Large' '4 Windows'	Value 'Small' 'Side by Side'	Value 'Medium' '3 Windows'
# PIP examples InterfaceName.Update('PIP') Value = InterfaceName.ReadStatus('PIP') InterfaceName.SubscribeStatus('PIP', None, FeedbackHandler)			
Command PIPIInput	Value 'VGA' 'HDMI 3' 'DisplayPort'	Value 'HDMI 1' 'HDMI 4' 'IPC/OPS'	Value 'HDMI 2' 'HDMI 5 (Front)' 'Media Player (Win/Android)'
# PIPIInput examples InterfaceName.Update('PIPIInput') Value = InterfaceName.ReadStatus('PIPIInput') InterfaceName.SubscribeStatus('PIPIInput', None, FeedbackHandler)			
Command PIPIInputWindow2	Value 'VGA' 'HDMI 3' 'DisplayPort'	Value 'HDMI 1' 'HDMI 4' 'IPC/OPS'	Value 'HDMI 2' 'HDMI 5 (Front)' 'Media Player (Win/Android)'
# PIPIInputWindow2 examples InterfaceName.Update('PIPIInputWindow2') Value = InterfaceName.ReadStatus('PIPIInputWindow2') InterfaceName.SubscribeStatus('PIPIInputWindow2', None, FeedbackHandler)			
Command PIPIInputWindow3	Value 'VGA' 'HDMI 3'	Value 'HDMI 1' 'HDMI 4'	Value 'HDMI 2' 'HDMI 5 (Front)'

	'DisplayPort'	'IPC/OPS'	'Media Player (Win/Android)'
# PIPInputWindow3 examples InterfaceName.Update('PIPInputWindow3') Value = InterfaceName.ReadStatus('PIPInputWindow3') InterfaceName.SubscribeStatus('PIPInputWindow3', None, FeedbackHandler)			
Command PIPPosition	Value 'Bottom Left' 'Top Right'	Value 'Bottom Right'	Value 'Top Left'
# PIPPosition examples InterfaceName.Update('PIPPosition') Value = InterfaceName.ReadStatus('PIPPosition') InterfaceName.SubscribeStatus('PIPPosition', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command Volume	Value 0 to 100 in steps of 1		
# Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

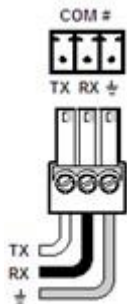
Data Bits: 8

Parity: None

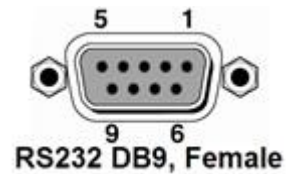
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	No
Multi-Connection Capabilities:	No
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 4:3	\x07\x01\x02ASP\x02\x08
Aspect Ratio Full Screen	\x07\x01\x02ASP\x01\x08
Aspect Ratio Letterbox	\x07\x01\x02ASP\x03\x08
Aspect Ratio Native	\x07\x01\x02ASP\x00\x08
Audio Mute Off	\x07\x01\x02MUT\x00\x08
Audio Mute On	\x07\x01\x02MUT\x01\x08
Auto Image None	\x07\x01\x02ADJ\x00\x08
Executive Mode Off	\x07\x01\x02KLC\x00\x08
Executive Mode On	\x07\x01\x02KLC\x01\x08
Input DisplayPort	\x07\x01\x02MIN\x0D\x08
Input HDMI 1	\x07\x01\x02MIN\x09\x08
Input HDMI 2	\x07\x01\x02MIN\x0A\x08
Input HDMI 3	\x07\x01\x02MIN\x0B\x08
Input HDMI 4	\x07\x01\x02MIN\x0C\x08
Input HDMI 5 (Front)	\x07\x01\x02MIN\x11\x08
Input IPC/OPS	\x07\x01\x02MIN\x0E\x08
Input Media Player (Win/Android)	\x07\x01\x02MIN\x12\x08
Input VGA	\x07\x01\x02MIN\x00\x08
Menu Navigation Down	\x07\x01\x02RCU\x03\x08
Menu Navigation Enter	\x07\x01\x02RCU\x06\x08
Menu Navigation Exit	\x07\x01\x02RCU\x07\x08
Menu Navigation Info	\x07\x01\x02RCU\x01\x08
Menu Navigation Left	\x07\x01\x02RCU\x04\x08
Menu Navigation Menu	\x07\x01\x02RCU\x00\x08
Menu Navigation Right	\x07\x01\x02RCU\x05\x08
Menu Navigation Up	\x07\x01\x02RCU\x02\x08
Picture Mode Cinema	\x07\x01\x02SCM\x03\x08
Picture Mode Game	\x07\x01\x02SCM\x02\x08
Picture Mode Sport	\x07\x01\x02SCM\x01\x08
Picture Mode User	\x07\x01\x02SCM\x00\x08
Picture Mode Vivid	\x07\x01\x02SCM\x04\x08
PIP 3 Windows	\x07\x01\x02PSC\x06\x08
PIP 4 Windows	\x07\x01\x02PSC\x07\x08
PIP Input DisplayPort	\x07\x01\x02PIN\x0D\x08
PIP Input HDMI 1	\x07\x01\x02PIN\x09\x08
PIP Input HDMI 2	\x07\x01\x02PIN\x0A\x08
PIP Input HDMI 3	\x07\x01\x02PIN\x0B\x08
PIP Input HDMI 4	\x07\x01\x02PIN\x0C\x08
PIP Input HDMI 5 (Front)	\x07\x01\x02PIN\x11\x08
PIP Input IPC/OPS	\x07\x01\x02PIN\x0E\x08
PIP Input Media Player (Win/Android)	\x07\x01\x02PIN\x12\x08
PIP Input VGA	\x07\x01\x02PIN\x00\x08

Global Scriptor Module Communication Sheet

Revision: 10/31/2017

PIP Input Window 2 DisplayPort	\x07\x01\x02PIO\x0D\x08
PIP Input Window 2 HDMI 1	\x07\x01\x02PIO\x09\x08
PIP Input Window 2 HDMI 2	\x07\x01\x02PIO\x0A\x08
PIP Input Window 2 HDMI 3	\x07\x01\x02PIO\x0B\x08
PIP Input Window 2 HDMI 4	\x07\x01\x02PIO\x0C\x08
PIP Input Window 2 HDMI 5 (Front)	\x07\x01\x02PIO\x11\x08
PIP Input Window 2 IPC/OPS	\x07\x01\x02PIO\x0E\x08
PIP Input Window 2 Media Player (Win/Android)	\x07\x01\x02PIO\x12\x08
PIP Input Window 2 VGA	\x07\x01\x02PIO\x00\x08
PIP Input Window 3 DisplayPort	\x07\x01\x02PIP\x0D\x08
PIP Input Window 3 HDMI 1	\x07\x01\x02PIP\x09\x08
PIP Input Window 3 HDMI 2	\x07\x01\x02PIP\x0A\x08
PIP Input Window 3 HDMI 3	\x07\x01\x02PIP\x0B\x08
PIP Input Window 3 HDMI 4	\x07\x01\x02PIP\x0C\x08
PIP Input Window 3 HDMI 5 (Front)	\x07\x01\x02PIP\x11\x08
PIP Input Window 3 IPC/OPS	\x07\x01\x02PIP\x0E\x08
PIP Input Window 3 Media Player (Win/Android)	\x07\x01\x02PIP\x12\x08
PIP Input Window 3 VGA	\x07\x01\x02PIP\x00\x08
PIP Large	\x07\x01\x02PSC\x03\x08
PIP Medium	\x07\x01\x02PSC\x02\x08
PIP Off	\x07\x01\x02PSC\x00\x08
PIP Position Bottom Left	\x07\x01\x02PPO\x00\x08
PIP Position Bottom Right	\x07\x01\x02PPO\x01\x08
PIP Position Top Left	\x07\x01\x02PPO\x02\x08
PIP Position Top Right	\x07\x01\x02PPO\x03\x08
PIP Side by Side	\x07\x01\x02PSC\x04\x08
PIP Small	\x07\x01\x02PSC\x01\x08
PIP Swap None	\x07\x01\x02SWA\x00\x08
Power Off	\x07\x01\x02POW\x00\x08
Power On	\x07\x01\x02POW\x01\x08
RC Emulation Blank Screen	\x07\x01\x02RCU+\x08
RC Emulation Freeze	\x07\x01\x02RCU\x18\x08
RC Emulation Volume Down	\x07\x01\x02RCU\x1E\x08
RC Emulation Volume Up	\x07\x01\x02RCU\x1D\x08
Volume 0	\x07\x01\x02VOL\x00\x08
Volume 100	\x07\x01\x02VOLd\x08

Appendix B. Query Commands

Aspect Ratio	\x07\x01\x01ASP\x08
Audio Mute	\x07\x01\x01MUT\x08
Executive Mode	\x07\x01\x01KLC\x08
Input	\x07\x01\x01MIN\x08
Picture Mode	\x07\x01\x01SCM\x08
PIP	\x07\x01\x01PSC\x08
PIP Input	\x07\x01\x01PIN\x08
PIP Input Window 2	\x07\x01\x01PIO\x08
PIP Input Window 3	\x07\x01\x01PIP\x08
PIP Position	\x07\x01\x01PPO\x08
Power	\x07\x01\x01POW\x08
Volume	\x07\x01\x01VOL\x08